

Kick Parameter Guide

Parameter reference — Thee Coast Machine

1. Internal Structure

The Pacific Synthesis kick is a three-layer synth: **sine oscillator + pitch envelope + click burst**. Modelled after TR-808 / 909 analog kick synthesis, with fully tweakable modern controls.

Element	Role	Sonic character
Sine oscillator	Main tone (body)	"Boom" / "Thump"
Pitch envelope	Drops quickly from high to low — defines the punch	The "thoom" sweep
Click burst	Very short noise transient (~2 ms)	Initial "tick" attack
Drive (tanh saturation)	Adds saturation and harmonics at output	Thicker / dirtier weight

2. Parameter Reference

PITCH (30 — 140 Hz)

Final pitch (= bass tone) of the kick.

- **30–45 Hz**: deep sub, almost felt (808-style, Dub, Trap)
- **45–60 Hz**: standard Techno / House (909-leaning, Detroit)
- **60–80 Hz**: punchy mid territory, less clashy with bass melody
- **80–140 Hz**: higher and tom-like (Industrial / IDM)

Tip: The lower you go, the more it depends on the playback system. Verify on a small speaker.

AMOUNT (0 — 8 (octaves))

Pitch envelope depth — how many octaves above PITCH the oscillator starts before dropping.

- **0**: no pitch motion, pure sine
- **1–3**: subtle drop (Lo-Fi / Dub)
- **3–5**: classic "thoom" (House / Techno)
- **5–7**: aggressive punch (909 / Industrial)
- **7–8**: extreme "zap"

Tip: PITCH and AMOUNT together define about 80% of the kick character.

DECAY (0.05 — 1.5 sec)

Sustain time of the sine envelope.

- **0.05–0.10**: very tight (140+ BPM Hard Techno)

- **0.15–0.25**: standard (House, mid-tempo Techno)
- **0.30–0.50**: sustained, sub-like (Trap, Dub)
- **0.60+**: long tail (808 melodic kick)

Tip: Long DECAY clashes with bass. Use sidechain (SC AMT) to duck the bass under the kick.

DRIVE (0 — 1)

Tanh soft saturation on the output.

- **0.0**: clean
- **0.1–0.3**: subtle harmonics, warmth
- **0.3–0.6**: thick, forward (modern Techno staple)
- **0.6–0.9**: obvious distortion (Industrial)
- **1.0**: fully distorted

Tip: DRIVE emphasizes the pitch envelope. High AMOUNT + high DRIVE = "thwack".

3. Character Presets

Goal	PITCH	AMOUNT	DECAY	DRIVE
808 deep sub (Dub, Trap)	42 Hz	4.5	0.55	0.12
909 punchy (House, Techno)	60 Hz	4.8	0.40	0.35
Classic Detroit	50 Hz	4.0	0.30	0.25
606 tight (Electro)	75 Hz	3.5	0.18	0.30
Hard Techno (140+)	55 Hz	5.5	0.18	0.65
Industrial / distorted	50 Hz	6.0	0.25	0.85
Lo-Fi / Boom Bap	52 Hz	4.0	0.30	0.25
Dub Sub (bass replacement)	40 Hz	3.0	0.80	0.10
Acid Tech (Hardfloor)	46 Hz	3.8	0.24	0.25
Ambient (soft)	55 Hz	2.5	0.35	0.05

4. Tuning Order

#	Do this	Why
Step 1	PITCH (low-end weight)	Decide first how low the kick should sit
Step 2	DECAY (length)	Driven by tempo and surroundings
Step 3	AMOUNT (punch)	Punchy or mellow?
Step 4	DRIVE (character)	Forward or clean?
Step 5	VOL & PAN	Always judge against the full mix

5. Troubleshooting

Symptom	Cause & fix
Kick feels "not there"	PITCH too low. Check on a small speaker, or raise DRIVE for harmonics.
Kick floats / sits oddly	PITCH may clash with another instrument. Nudge ± 5 Hz.
Kick and bass mud together	Raise SC AMT (sidechain) to 0.3–0.5 to duck the bass.
Weak attack	Raise AMOUNT, or DRIVE above 0.3.
Too distorted	Lower DRIVE, or lower VOL (limiter may be reacting).
Loudness mismatch	Short DECAY = lower RMS. Use longer DECAY + DRIVE for level.

※ The kick is the mix reference. Lock it in first; placement of other voices becomes easier. Always tune in context, not in isolation.